

Review CSC1-338

Sets: A set is an unordered collection of distinct objects.

$$V = \{A, E, I, O, U\}$$

$$\text{bits} = \{0, 1\}$$

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

Size of the set S using $|S|$, this denotes how many elements we have in the set.

$$|V| = 5 \quad |\text{bits}| = 2$$

$$A \in V \quad \leftarrow \quad A \text{ exists in } V$$

$$B \notin V \quad \leftarrow \quad B \text{ does not exist in } V$$

Subsets

A set A is a subset of set B , if every element in set A appears in set B .

we denote this $A \subseteq B$ ← subset

If A is not a subset of B , we denote this using

$$A \not\subseteq B$$

$$A = \{2, 5\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$A \subseteq B \quad \checkmark$$

$$B \subseteq A \quad \times$$

$$D = \{1, 6\}$$

$$D \not\subseteq B$$

Question: If $|A|=n$, how many subsets does A have?

$$2^n$$

$A = \{1, 2, 3\}$, what is the powerset of A

$$P(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

$$|P(A)| = 8 = 2^3$$

Def: A is a proper subset of B , if $A \subseteq B$ and $|A| \neq |B|$, denoted as

$$A \subset B \text{ or } A \subsetneq B$$

$$A = \{1, 2\}, B = \{1, 2, 3, 4\}, D = \{1, 2, 3, 4\}$$

$$A \subset B \checkmark \quad B \subseteq D$$

Question, Given two sets X, Y

$$(X \subseteq Y) \wedge (Y \subseteq X) \Rightarrow X = Y$$

$$a, b \in \mathbb{R}$$

$$(a \leq b \wedge b \leq a) \Rightarrow a = b$$

Multisets: A set that allows duplicate elements.

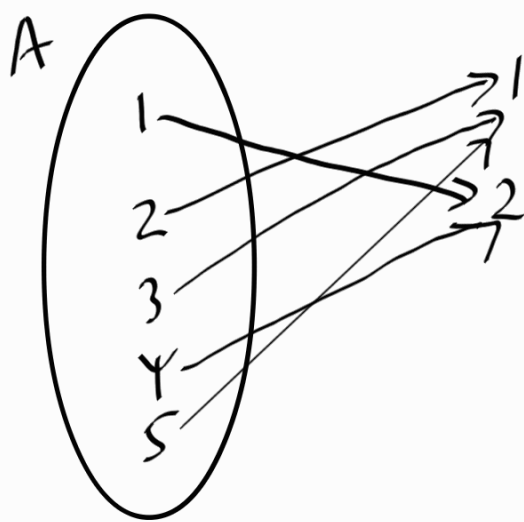
$$M = \{1, 1, 2, 3, 4, 4, 5\}$$

← multiset
← not a set.

A multiset M is an ordered pair (A, m) where A is the underlying set, and m is a function such that $m: A \rightarrow \mathbb{Z}^+$ that denotes multiplicity.

$$M = \{1, 2, 1, 3, 4, 5, 4\}$$

$$A = \{1, 2, 3, 4, 5\}$$



$$m(1) = 2$$

$$m(2) = 1$$

$$m(3) = 1$$

$$m(4) = 2$$

$$m(5) = 1$$

Some important Sets

$\mathbb{Z}$ = the set of all integers

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$$

$\mathbb{N}$ = The set of natural numbers

$$\mathbb{N} = \{ 1, 2, 3, \dots \}$$

$$\mathbb{Z}^+ = \{ 1, 2, 3, 4, \dots \}$$

$$\mathbb{Z}^- = \{ \dots, -2, -1 \}$$

$$\mathbb{Z} = \mathbb{Z}^+ \cup \mathbb{Z}^- \cup \{0\}$$

$\mathbb{R}$ = The set of all real numbers

$$\mathbb{Z}^{\geq 0} = \{ 0, 1, 2, \dots \}$$

$\mathbb{Q}$ = The set of rational numbers

The set builder notation

$$S = \{ x : \text{rule about } x \}$$

Ex:

$$\mathbb{Q} = \left\{ a : (p, q, \in \mathbb{Z}) \wedge (q \neq 0) \wedge (a = \frac{p}{q}) \right\}$$

$$\text{EVENs} = \{ e \in \mathbb{Z} : e \bmod 2 = 0 \}$$

Set operations

1. Union ($\cup$)

2. Intersection ($\cap$)

3. Complement ($\bar{}$)

4. Exclusive or ($\oplus$)

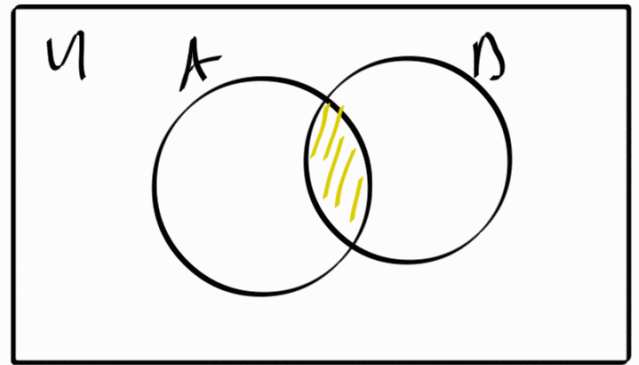
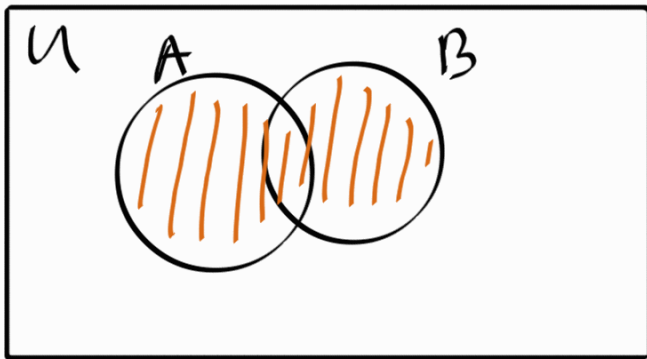
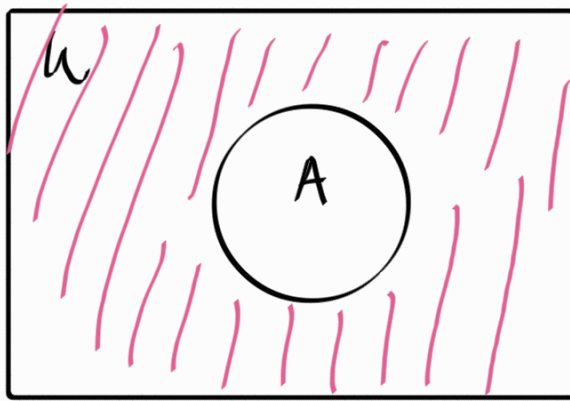
$$A = \{1, 3, 5, 6\} \quad B = \{2, 4, 6, 8\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$$

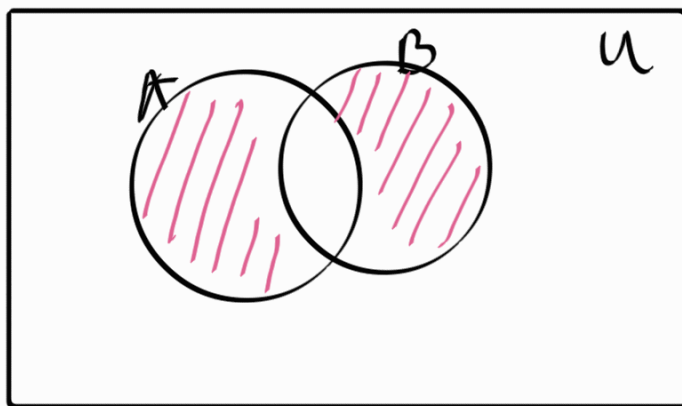
$$A \cap B = \{6\}$$

$$\bar{A} = \{2, 4, 7, 8\}$$

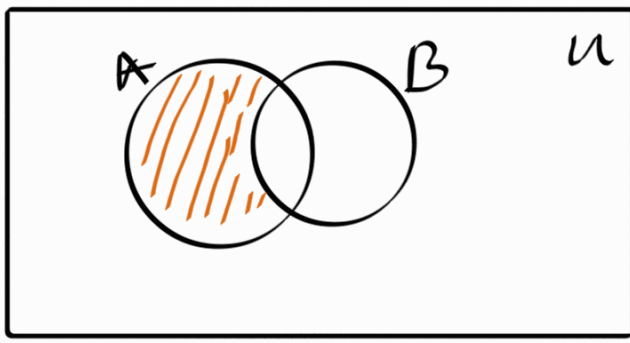


$$A = \{1, 3, 5, 6\} \quad B = \{2, 4, 6, 8\}$$

$$A \oplus B = \{1, 3, 5, 2, 4, 8\}$$



$$A - B = A \setminus B = \{1, 3, 5\}$$



$$A - B \neq B - A, \text{ when } A \neq B$$

Note:

$$A \cup \bar{A} = U$$

Sequences and Tuples.

- A sequence is a list of ordered elements.

$$\langle 1, 5, 6, 7 \rangle = (1, 5, 6, 7)$$

$$\langle 1, 5, 6, 7 \rangle \neq \langle 1, 6, 5, 7 \rangle$$

- finite sequences are called tuples
- 2-tuple is called a pair.

$$(a, b), \langle a, b \rangle$$

ex: A graph $G = (V, E)$

- V is the set of vertices
- E is the set of edges